

# Changes in flow-mediated brachial artery vasoactivity with lowering of desirable cholesterol levels in healthy middle-aged men.

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Current National Cholesterol Education Program guidelines consider desirable total and low-density lipoprotein cholesterol levels to be < 200 and < 160 mg/dl, respectively, for healthy individuals without multiple coronary risk factors. To determine the extent to which these levels affect vascular function, we assessed flow-mediated (endothelium-dependent) brachial artery vasoactivity noninvasively before, during, and after cholesterol lowering (simvastatin 10 mg/day) in 7 healthy middle-aged men with cholesterol levels meeting current recommendations. Flow-mediated brachial artery vasoactivity was measured using 7.5 MHz ultrasound and expressed as percent diameter change from baseline to hyperemic conditions (1 minute following 5 minutes of blood pressure cuff arterial occlusion). Flow-mediated vasoactivity rose from 5.0  $\pm$  3.6% at baseline to 10.5  $\pm$  5.6%, 13.3  $\pm$  4.3%, and 15.7  $\pm$  4.9% (all  $p < 0.05$ ) as cholesterol fell from 200  $\pm$  12 to 161  $\pm$  18, 169  $\pm$  16, and 153  $\pm$  11 mg/dl after 2, 4, and 12 weeks, respectively, of cholesterol-lowering therapy. Vasoactivity and cholesterol returned to baseline levels 12 weeks after simvastatin discontinuation. Overall, vasoactivity was found to correlate inversely with cholesterol levels ( $r = -0.47$ ,  $p = 0.004$ ). These data suggest that flow-mediated brachial artery vasoactivity responds rapidly to changes in cholesterol levels and that endothelial function improves by lowering cholesterol levels below recommendations of current guidelines.